

Indian Statistical Institute
B. Math. Hons. I Year
Semestral Examination 2002-2003
Discrete Mathematics

Date: 28-04-2003

Marks: 100

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Note: Answer questions upto a maximum of 100 marks.

1. (a) Let $f(x) \in F_q[x]$ be an irreducible polynomial and let α be a root of $f(x)$ in an extension field of F_q . Show that α^q also is a root of $f(x)$. Deduce that, if the degree of $f(x)$ is n , then $\alpha, \alpha^q, \dots, \alpha^{q^{n-1}}$ are all *distinct* roots of $f(x)$. [6+6]
(b) Show that in a finite field, any element can be written as a sum of two squares. [6]
(c) If F is a field such that each polynomial with coefficients in F has a root in F , show that F is infinite. [3]
2. (a) Define a projective plane of order n . Show that a projective plane of order 2 is unique. [2+5]
(b) Let S be a set of $n+2$ points in a projective plane of order n such that any line meets S in at most two points. How many lines of the plane meet in:
(i) one point of S ;
(ii) two points of S ;
(iii) no points of S . [3+3+3]
3. (a) Let C be the binary code associated with a projective plane (P, L) of order n . If $x \in F_2^P$ is a word in $C^\perp$ of weight $n+2$, show that any line meets the support of x in 0 or 2 points.
(b) Define a q -ary Hamming code H . Determine the weight enumerator of its dual code.
(c) Define a cyclic code of length n over a field F_q . Determine the parameters of the cyclic code of length 7 generated by the polynomial $x^2 + x + 1$. [8+8+8]

4. (a) Let G be the group of all nonsingular matrices with determinant 1 and the entries from F_5 . Let Z be the subgroup of G consisting of elements which commutes with each element of G . Determine:
 - (i) the elements of Z
 - (ii) order of the factor group G/Z . [8+8]
 (b) Determine the integer t such that G/Z is t -transitive, but not $(t+1)$ -transitive, on the set of 1-dimensional subspaces of a vector space of dimension 2 over F_q . [6]
5. (a) Prove that $1 + \frac{1}{2} + \dots + \frac{1}{n}$ is not an integer for any $n > 1$.
 (b) Show that, for positive integers $m > n$, $(2^{2^n} + 1, 2^{2^m} + 1) = 1$. Deduce that the number of primes is infinite.
 (c) Find an integer x such that $2x \equiv 1 \pmod{3}$, $3x \equiv 1 \pmod{5}$ and $5x \equiv 1 \pmod{7}$. [6+8+6]
6. (a) Define the quadratic residue of an integer modulo an odd prime integer p . Determine the quadratic nonresidues modulo the prime p . State precisely Gauss criteria to determine if a nonzero integer is a quadratic residue modulo p . [3+5+4]
 (b) State precisely the quadratic law of reciprocity. Determine the odd primes p for which:
 - (i) 2 is a quadratic residue modulo p ;
 - (ii) 5 is a quadratic residue modulo p . [2+4+4]